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A GEOMETRIC FRAMEWORK FOR REACTION-WHEEL OPERATING LIMITS IN GN&C

Anubhav Gupta*

Reaction-wheel performance is constrained by inverter current limits, voltage limits, and the electromechanical properties of the underlying permanent-magnet machine drive modeled in the synchronous dq frame. These constraints are often examined through time-domain simulations or torque–speed plots, which provide detailed results but do not always expose the geometric structure of the electrical operating limits. This paper presents a geometric visualization framework that unifies the inverter current circle, voltage ellipse, and constant-torque loci in the dq-plane, yielding a compact representation of reaction-wheel operating constraints. The framework illustrates how the geometry of current and voltage loci evolves with electrical speed, highlighting the transition between current-dominated and voltage-dominated regimes, the geometric manifestation of field weakening, and the resulting contraction of the torque–speed envelope at high speed. By relating regions of the (i_d, i_q) plane to torque, wheel speed, and stored angular momentum, the construction provides an interpretive complement to conventional simulation-based GN&C analysis, illustrated through a representative example.

INTRODUCTION

The classical circle diagram is a long-standing graphical method for visualizing the steady-state behavior of induction machines using only no-load and blocked-rotor test data.[1,2] By tracing the locus of stator current in the complex plane, the diagram provides a compact representation of how torque, slip, and power factor vary across the operating range, revealing structural relationships that are not always apparent in purely numerical simulations.

A broader treatment of classical machine testing and related geometric concepts is provided in Gupta.[3] An analytic–geometric reconstruction of the classical circle diagram, including a formulation based on Möbius transformations, is presented in,[4] while[5] introduces a computational framework for generating such diagrams. The present work does not revisit these derivations; rather, it draws inspiration from the geometric viewpoint embodied by the classical circle diagram and applies it to a different class of machines and applications.

A similar need for compact and interpretable visualizations arises in spacecraft attitude control, particularly for reaction wheel actuators. Reaction-wheel performance is fundamentally constrained by inverter current limits, voltage limits, and motor electromechanical properties. These constraints are commonly examined through torque–speed curves or time-domain actuator simulations, which provide detailed information but do not always emphasize the geometric structure of the electrical operating space as wheel speed and bus voltage vary.

In this work, the geometric perspective of the classical circle diagram is adapted to reaction-wheel drives based on permanent-magnet synchronous machines (PMSMs). Using the standard PMSM model expressed in the dq reference frame, the current constraint is shown to form a circle,

*Guidance, Navigation, and Control (GN&C) Engineer, In Orbit Aerospace Inc., Torrance, CA 90501.

while the voltage constraint forms an ellipse whose size and position depend on electrical speed. The intersection of these loci defines a geometrically bounded region in the (i_d, i_q) plane that may be interpreted in terms of combinations of torque, wheel speed, and angular momentum.

By presenting the current and voltage loci together with constant-torque lines, a unified graphical picture emerges that distinguishes current-limited and voltage-limited regimes, illustrates the role of field weakening, and highlights the onset of torque saturation. This geometric construction serves as an intuitive complement to conventional simulation-based analyses, clarifying how electrical operating constraints reshape the space of possible operating points as electrical speed and system parameters vary.

The contribution of this work is a geometric framework for visualizing reaction-wheel operating limits that parallels the classical circle diagram while being formulated specifically for PMSM-based spacecraft actuators. The resulting diagram is intended to support design-level analysis and educational understanding in spacecraft guidance, navigation, and control.

GEOMETRIC FRAMEWORK FOR PMSM ELECTRICAL CONSTRAINTS

While flight reaction wheels may employ BLDC or PMSM hardware, both are commonly represented using an equivalent permanent-magnet synchronous machine model in the dq reference frame for steady-state analysis. The flux linkages are

$$\psi_d = L_d i_d + \psi_m, \tag{1}$$

$$\psi_q = L_q i_q, \tag{2}$$

where L_d and L_q are the direct- and quadrature-axis inductances and ψ_m is the permanent-magnet flux linkage implied by the motor's torque constant. The corresponding stator voltages are

$$v_d = R_s i_d - \omega_e \psi_q, \tag{3}$$

$$v_q = R_s i_q + \omega_e \psi_d. \tag{4}$$

The inverter imposes two constraints: a current limit and a voltage (bus) limit,

$$i_d^2 + i_q^2 \leq I_{\max}^2, \tag{5}$$

$$v_d^2 + v_q^2 \leq V_{\max}^2. \tag{6}$$

Substituting the voltage expressions into the second constraint yields

$$(R_s i_d - \omega_e L_q i_q)^2 + (R_s i_q + \omega_e L_d i_d + \omega_e \psi_m)^2 \leq V_{\max}^2. \tag{7}$$

For high-quality reaction wheels, R_s is typically small. Neglecting R_s is appropriate when $\omega_e L \gg R_s$ (i.e., above low electrical frequencies for typical reaction-wheel-class windings), which yields the simplified steady-state constraint

$$(\omega_e L_d i_d + \omega_e \psi_m)^2 + (\omega_e L_q i_q)^2 \leq V_{\max}^2. \tag{8}$$

Dividing both terms by $V_{\max}^2$ produces the voltage ellipse

$$\frac{(i_d + \psi_m/L_d)^2}{(V_{\max}/\omega_e L_d)^2} + \frac{i_q^2}{(V_{\max}/\omega_e L_q)^2} = 1, \tag{9}$$

centered at $(-\psi_m/L_d,\, 0)$ with semi-axes

$$a = \frac{V_{\max}}{\omega_e L_d}, \qquad b = \frac{V_{\max}}{\omega_e L_q}. \tag{10}$$

As speed increases, both axes shrink, corresponding to the familiar field-weakening effect. When $L_d = L_q = L$ (surface-mounted PMSM), Eq. (9) reduces to the circle

$$(i_d + \psi_m/L)^2 + i_q^2 = (V_{\max}/\omega_e L)^2. \tag{11}$$

The effective voltage limit depends on inverter modulation:

$$V_{\max} \approx \frac{m_{\max} V_{\text{bus}}}{\sqrt{3}} \quad \text{(SPWM)}, \tag{12}$$

$$V_{\max} \approx \frac{V_{\text{bus}}}{\sqrt{3}} \qquad \text{(SVPWM)}. \tag{13}$$

$$\tag{14}$$

Geometric Interpretation for Reaction Wheels

The current and voltage limits in the dq-plane define two geometric loci: a circle of radius $I_{\max}$ centered at the origin, and a voltage ellipse whose center and axes vary with electrical speed ω_e. The resulting region in the dq-plane is given by the intersection of these two sets and represents the portion of current space enclosed by the plotted electrical constraint loci.

Figure 1 illustrates the case $L_d \neq L_q$. The voltage ellipse shifts left by ψ_m/L_d and its axes shrink as ω_e increases. At low speed, the ellipse is large, and the current limit dominates, so the wheel is *current-limited*. The entire horizontal chord corresponding to a constant i_q lies inside the overlap of the current and voltage loci, illustrating how that torque level is positioned relative to the electrical constraint geometry at that speed.

As speed increases, the voltage ellipse contracts and eventually cuts into the left half-plane. This marks the transition to *voltage-limited* operation: only a portion of the torque line intersects the region defined by the plotted voltage and current loci. The range of torque levels lying within the plotted constraint geometry is therefore reduced, and negative i_d (field-weakening) operation becomes increasingly relevant. At sufficiently high speed, the torque line lies entirely outside the voltage ellipse, illustrating the geometric conditions under which torque saturation arises as electrical speed increases.

Figure 2 shows the surface-mounted PMSM case $L_d \approx L_q$, where the voltage locus becomes a circle. The geometric picture remains identical—at low speed the voltage circle encompasses the current circle, while at high speed it shrinks beneath it—but the symmetry simplifies interpretation.

The geometric diagram therefore provides at a glance:

- the distinction between current-limited and voltage-limited regimes,
- the role of field-weakening as electrical speed increases,
- the geometric relationship between (i_d, i_q) space and constant-torque loci,

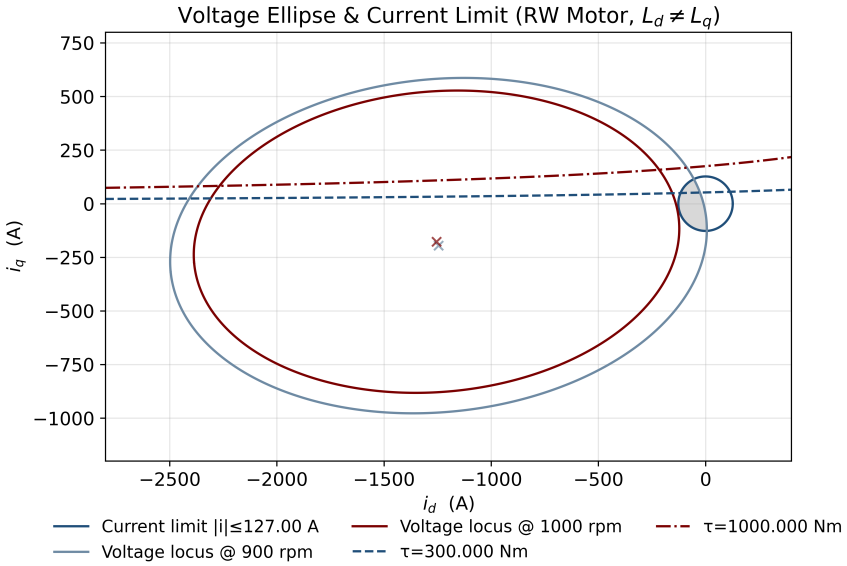


Figure 1 dq**-plane voltage ellipse for** $L_d \neq L_q$**. The ellipse shrinks and shifts left with speed; the shaded region shows overlap with the current circle. Example constant-torque loci are shown for illustration; only the portion lying within the overlap is electrically admissible at the given speed.**

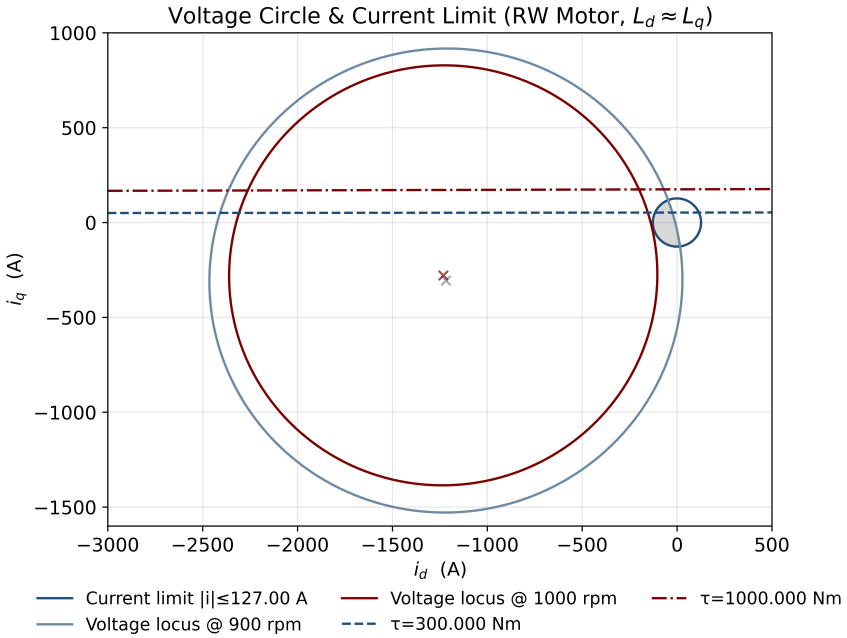


Figure 2 dq**-plane voltage circle for** $L_d \approx L_q$**. The circle shrinks and shifts left with speed; the shaded region indicates overlap with the current circle at low speed (electrically admissible region).**

- the speed-dependent geometric structure underlying the torque–speed envelope,

- the geometric conditions associated with torque saturation.

This visual interpretation parallels the classical Heyland diagram for induction machines: a single static geometric construction encodes the global operating structure that would otherwise require extensive numerical simulation to reveal.

Mapping *dq* Currents to Torque, Speed, and Momentum

For spacecraft GN&C applications, the dq-plane diagram provides an interpretive visualization linking electrical operating loci to torque, wheel speed, and stored angular momentum. The diagram does not evaluate command admissibility or enforce operating limits; rather, it illustrates how electrical constraints shape the geometry of possible operating points across the reaction wheel operating envelope.

Electromagnetic torque is given by

$$\tau = \frac{3}{2} p(\psi_d i_q - \psi_q i_d) \tag{15}$$

$$= \frac{3}{2} p\left(\psi_m i_q + (L_d - L_q) i_d i_q\right). \tag{16}$$

For surface-mounted machines ($L_d \approx L_q$), torque simplifies to

$$\tau = \frac{3}{2} p\, \psi_m\, i_q, \tag{17}$$

so constant torque corresponds to horizontal lines in the dq-plane (fixed i_q). Wheel speed relates to electrical speed via

$$\omega_e = p\, \frac{2\pi\Omega}{60}, \tag{18}$$

and the stored angular momentum is

$$\mathbf{H} = \mathbf{J}\, \omega_{\text{mech}} = \mathbf{J}\, \frac{2\pi\Omega}{60}. \tag{19}$$

Connection to Reaction-Wheel Torque–Speed and Momentum

Using the mappings defined in the previous subsection, each point (i_d, i_q) in the dq-plane may be interpreted in terms of reaction-wheel actuator quantities relevant to spacecraft GN&C:

- **Torque:** For surface-mounted machines ($L_d \approx L_q$), electromagnetic torque depends primarily on the quadrature current i_q. Constant-torque commands, therefore, appear as horizontal lines in the dq-plane. The portion of a torque line that lies within the intersection of the current and voltage loci represents electrically admissible operating points at the given wheel speed.
- **Speed:** Electrical speed increases linearly with mechanical wheel speed, causing the voltage ellipse to contract with increasing Ω. This provides a geometric visualization of the transition from current-limited to voltage-limited operation and the onset of torque saturation at high speed.

- **Momentum Storage:** Stored angular momentum increases with wheel speed. Geometrically, this may be interpreted by examining how constant-torque loci are positioned relative to the voltage and current constraints as speed increases, illustrating how momentum accumulation progressively constrains admissible operating points.

- **Torque–Speed Envelope:** The classical torque–speed envelope emerges naturally from the geometry as the voltage locus contracts with speed, reducing the extent of constant-torque loci intersecting the admissible region and producing the familiar high-speed torque roll-off.

Geometric Transition Between Constraint-Dominated Regimes

In the surface-mounted PM (SPM) case where $L_d = L_q = L$, the voltage constraint in the dq-plane reduces to a circle. For electrical speed ω_e, the voltage locus is a circle of radius

$$R_v(\omega_e) = \frac{V_{\max}}{\omega_e L}, \tag{20}$$

whose center is located at

$$(i_d, i_q) = \left(-\frac{\psi_m}{L},\, 0\right). \tag{21}$$

The current constraint is also a circle centered at the origin with radius $R_i = I_{\max}$.

At low speed, the voltage circle contains the current circle and the machine is current-limited. As speed increases, the voltage circle shrinks, and the geometric overlap becomes constrained by the voltage limit. The base speed corresponds to the speed at which the two circles are tangent. As electrical speed increases, the voltage locus contracts relative to the current constraint. At a particular operating speed, the two loci become externally tangent, marking a geometric transition between current-dominated and voltage-dominated regions of the dq-plane. This transition corresponds to the familiar base-speed concept in PMSM operation and is included here solely as a geometric observation rather than as a design criterion, control boundary, or operational limit.

Torque Lines and Null-Space Interpretation

For $L_d \approx L_q$, the torque expression is independent of i_d, so i_d spans a null-space direction of Eq. (16). In practice this null-space is truncated by the current circle and voltage ellipse, so only the segment lying inside the plotted constraint geometry corresponds to operating points illustrated within the voltage and current loci. For salient machines ($L_d \neq L_q$), torque lines tilt because of the cross-term $(L_d - L_q)i_d i_q$, and negative i_d can be used to achieve maximum torque-per-ampere (MTPA).

ILLUSTRATIVE EXAMPLE

The reaction wheels considered in this work employ PMSM motors whose parameters are often proprietary; however, representative values can be drawn from the automotive traction-motor literature. For illustrative purposes, we adopt the PMSM parameters reported in Feng et al.,[6] summarized in Table 1. The numerical values are chosen to clearly illustrate the geometric behavior of the proposed framework rather than to represent a specific flight-qualified reaction wheel. While the numerical example uses parameters drawn from the automotive PMSM literature, the resulting

Table 1 Electrical parameters for representative PMSM.

Parameter	Description	Value	Unit
p	Pole pairs	6	–
ψ_m	Flux linkage	0.635	Wb
R_s	Stator resistance	0.071	Ω
L_d	d-axis inductance	0.49	mH
L_q	q-axis inductance	0.50	mH
$I_{\max}$	Current limit	127	A
V_{dc}	DC bus voltage	650	V

geometric structure depends only on standard permanent-magnet machine parameters and applies equally to reaction-wheel actuators modeled in the synchronous dq frame.

Reaction wheels span a wide range of sizes and electrical interfaces, with spacecraft bus voltages commonly in the 28–50 V class and actuator currents typically in the ampere range, scaling with wheel size and performance class. Reaction-wheel motors are brushless machines driven by PWM inverters, and their electrical characteristics (e.g., L in the mH range and R_s in the sub-ohm to ohm range) are consistent with small three-phase machines modeled in prior literature.[7] The numerical parameter set in Table 1 is therefore used here primarily to clearly separate the voltage and current loci in the plots; the geometric structure and interpretations are unchanged under scaling to flight-level buses and current limits.

This construction is visually reminiscent of the classical circle diagram used in induction-machine analysis, as well as analytic reconstructions based on Möbius mappings.[4] However, the present diagram is formulated directly in the PMSM dq frame for reaction-wheel actuators and characterizes electrical operating constraints arising from current and voltage limits, rather than slip–power factor relationships. The resulting geometric object is therefore distinct, both analytically and in scope, from classical induction-machine circle-diagram formulations.

Note. The i_d axis range reflects the negative current required to fully demagnetize the permanent-magnet field. Although these values exceed typical actuator current limits, they are shown to correctly scale the voltage-locus geometry. Only the intersection with the $\|i\| \leq I_{\max}$ region corresponds to physically admissible operating points.

The parameter set in Table 1 is drawn from the PMSM literature and is used to illustrate the geometry with clearly separated loci. Typical reaction-wheel-class BLDC/PMSM drives operate at lower bus voltages and currents, but the dq-plane construction depends only on standard machine parameters and preserves its geometric interpretation under scaling of $V_{\max}$ and $I_{\max}$.

Figure 3 shows the current-limit circle and voltage loci at two operating speeds for this motor. The voltage limit is taken from Eq. (13), consistent with SPWM and a small margin on the modulation index. At $\Omega = 900$ rpm, the voltage ellipse overlaps the current circle, and the shaded region indicates the geometric overlap of the plotted voltage and current loci. At $\Omega = 1000$ rpm, the voltage ellipse has contracted and shifted such that the overlap between the plotted loci vanishes for the chosen $V_{\max}$ and $I_{\max}$ (i.e., the admissible region becomes empty in this simplified steady-state model).

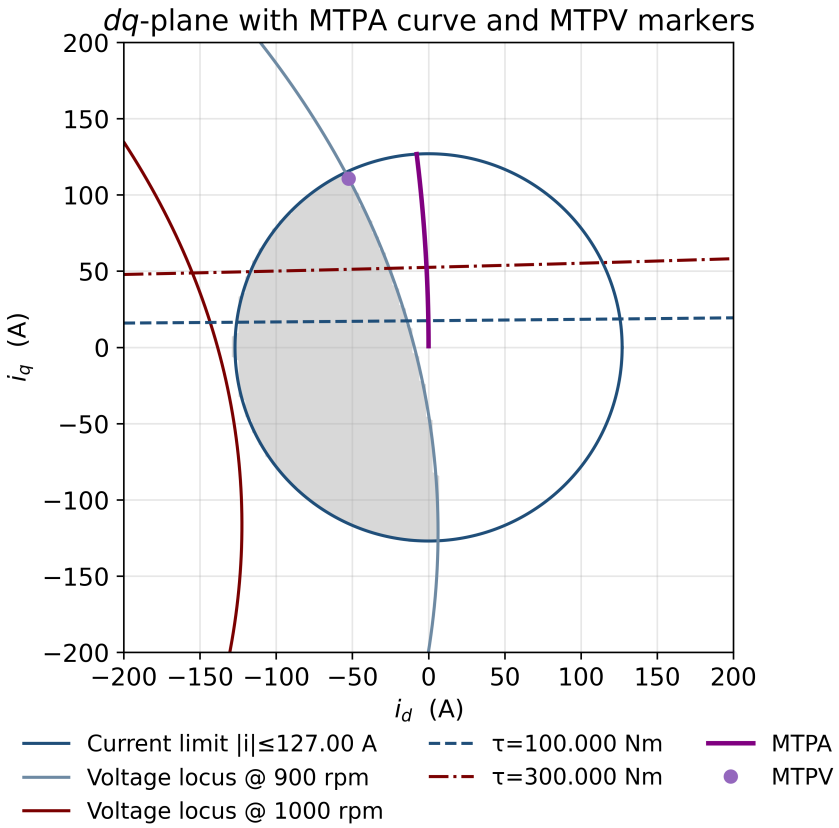


Figure 3 Current circle and voltage loci at two motor speeds for the PMSM in Table 1. The shaded region at 900 rpm indicates the geometric overlap of the plotted voltage and current loci; at 1000 rpm the loci are separated. Markers indicate MTPA and MTPV operating points. These points are shown for illustrative comparison only and are not proposed here as command-selection or control policies.

These observations are consistent with the geometric transition between current-dominated and voltage-dominated regimes described earlier. Substituting $p = 6$, $L \approx 0.495$ mH, $\psi_m = 0.635$ Wb, and $V_{\max} \approx 0.95\, V_{dc}/\sqrt{3}$ yields a characteristic electrical-speed scale of approximately $\Omega \approx 1000$ rpm for the illustrated geometry. The 900 rpm case, therefore, lies in a regime where current and voltage loci overlap, while the 1000 rpm case lies in a regime where the loci are separated.

To illustrate torque levels within this geometric picture, consider three representative constant-torque commands (shown here only to demonstrate how horizontal torque loci intersect the admissible region). For the nearly surface-mounted PMSM ($L_d \approx L_q$), the torque expression reduces to Eq. (17), so torque depends primarily on i_q and constant-torque lines appear as horizontal lines in the dq-plane. Using the parameters of Table 1, the corresponding quadrature currents for the three illustrative torque levels are approximately 17.5 A, 35 A, and 52.5 A, respectively. At 900 rpm, all three torque lines intersect the shaded region, illustrating their relative placement within the plotted constraint geometry at that operating speed.

At 1000 rpm, the absence of overlap between the current circle and voltage ellipse illustrates that these torque lines lie outside the geometric overlap of the current and voltage loci at the given bus

voltage. This example provides a static sensitivity analysis of the constraint geometry. It demonstrates how varying a speed parameter in the model alters the visualization of the electrical loci, serving as a tool for offline actuator selection and sizing rather than real-time operational assessment.

CONCLUSION

Reaction-wheel actuators are commonly characterized using numerical time-domain simulations or torque–speed curves, which can obscure the geometric structure of electrical operating constraints as electrical speed, bus voltage, and current limits vary. This paper presented a geometric framework that brings together the inverter current circle, voltage ellipse, and constant-torque loci in the dq-plane, providing a compact geometric visualization of reaction-wheel operating limits.

The framework illustrates the transition between current-limited and voltage-limited regimes, the emergence of field-weakening behavior, and the onset of torque saturation at high electrical speed. By relating regions of the (i_d, i_q) plane to torque, wheel speed, and stored angular momentum, the diagram offers an intuitive complement to conventional simulation-based analyses and supports insight into actuator behavior across the operating envelope.

The construction is conceptually rooted in classical machine-analysis diagrams, but is formulated here specifically for PMSM-based spacecraft reaction-wheel actuators. Potential extensions of the framework include the incorporation of thermal effects, parameter variation with aging, efficiency contours, and the generation of torque–speed envelopes for design-oriented GN&C studies. Accordingly, the framework is intended as a descriptive and interpretive tool, rather than as a mechanism for torque command generation or enforcement.

Dynamic inverter effects (e.g., current-loop bandwidth, PWM ripple) and thermal derating are not modeled; the framework is intended for steady-state, design-level interpretation of electrical operating limits. The framework is presented solely as a geometric visualization and interpretive aid. It does not evaluate torque-command admissibility, modify commanded torques, perform feasibility enforcement, or form part of a control law, allocator, or onboard decision-making algorithm.

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